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Delayed Weight Updates for High Throughput Hardware Accelerator of the CCSDS-123.0-B-2 Compression Algorithm

Master's thesis in Electronic System Design and Innovation
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NTNU

Abstract

Sammendrag

AI Declaration

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Chapter 1

Introduction

Chapter 2

Background

Test

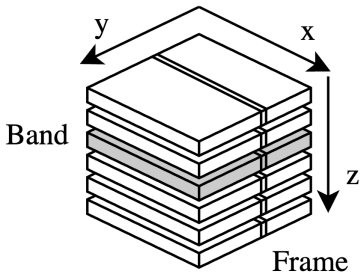

Figure 2.1

of

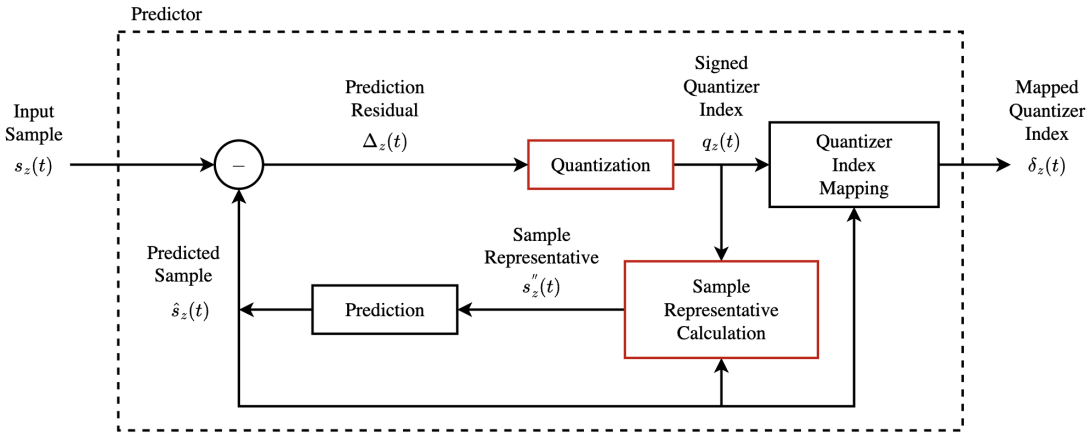
Predictor
Input Sample
$s_z(t)$
Prediction Residual
$\Delta_z(t)$
Quantization
Signed Quantizer Index
$q_z(t)$
Quantizer Index Mapping
Mapped Quantizer Index
$\delta_z(t)$
Sample Representative
$s''_z(t)$
Sample Representative Calculation
Prediction
Predicted Sample
$\hat{s}_z(t)$

Figure 2.2

multipage

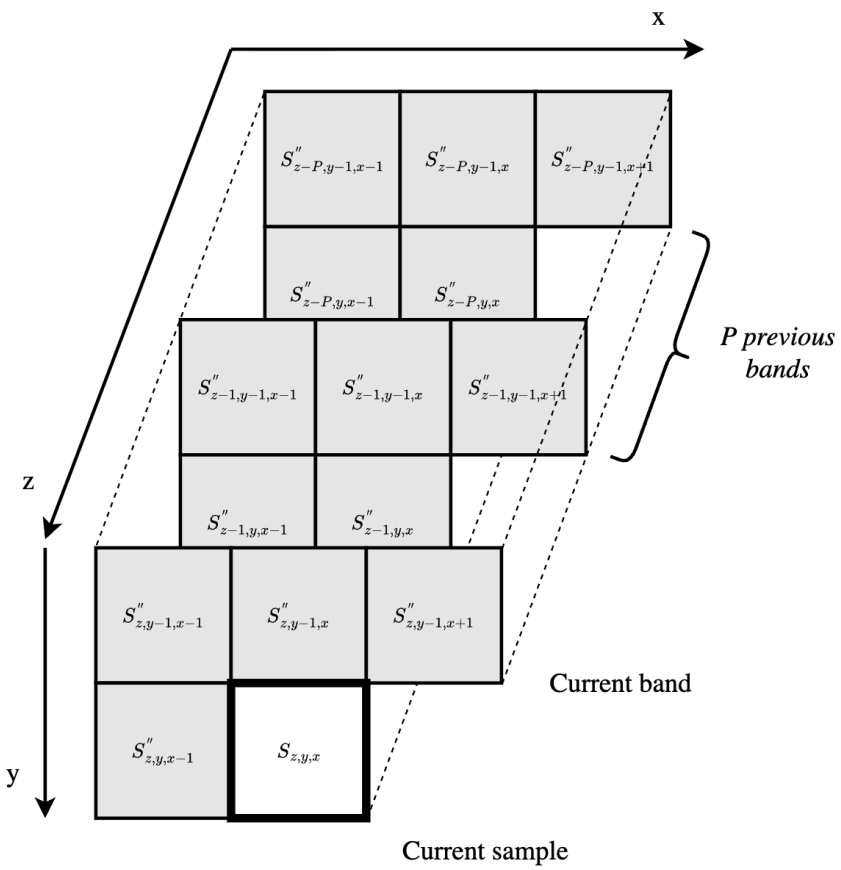

Figure 2.3

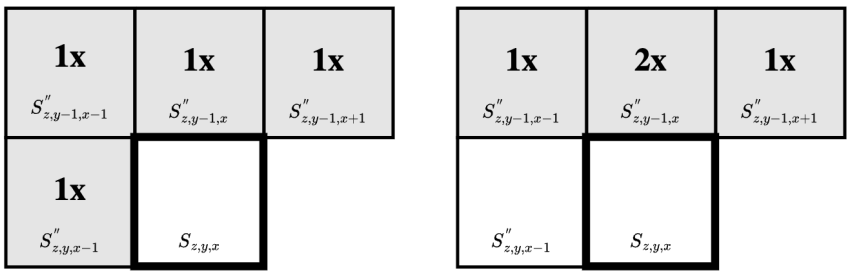

Figure 2.4

$$\sigma_{z,y,x} = \begin{cases} s''_{z,\,y-1,\,x-1} + 2s''_{z,\,y-1,\,x} + s''_{z,\,y-1,\,x+1}, & y > 0,\ 0 < x < N_X - 1, \\ 4s''_{z-1,\,y,\,x-1}, & y = 0,\ x > 0,\ z > 0, \\ 2\left(s''_{z,\,y-1,\,x} + s''_{z,\,y-1,\,x+1}\right), & y > 0,\ x = 0, \\ 2\left(s''_{z,\,y-1,\,x-1} + s''_{z,\,y-1,\,x}\right), & y > 0,\ x = N_X - 1, \\ 4s_{\text{mid}}, & y = 0,\ x = 0,\ z = 0. \end{cases} \tag{2.1}$$

$$\mathbf{U}_z(t) = \begin{bmatrix} d_{z-1}(t) \\ d_{z-2}(t) \\ \vdots \\ d_{z-P_z^*}(t) \end{bmatrix}, \tag{2.2}$$

$$P_z^* = \min(z, P), \tag{2.3}$$

$$\hat{d}_z(t) = \mathbf{W}_z(t) \cdot \mathbf{U}_z(t). \tag{2.4}$$

$$\mathbf{W}_z(t) = \begin{bmatrix} \omega_z^{(1)}(t) \\ \omega_z^{(2)}(t) \\ \vdots \\ \omega_z^{(P_z^*)}(t) \end{bmatrix}, \tag{2.5}$$

$$\omega_z^{(1)}(1) = \frac{7}{8} 2^{\Omega}, \qquad \omega_z^{(i)}(1) = \left\lfloor \frac{1}{8}\, \omega_z^{(i-1)}(1) \right\rfloor, \quad i = 2, 3, \ldots, P_z^*, \tag{2.6}$$

$$\omega_{\min} = -2^{\Omega+2},\ \omega_{\max} = -2^{\Omega+2} - 1. \tag{2.7}$$

$$\omega_{z,\text{offset}}^{(i)}(t) = \left\lfloor \frac{1}{2} \left(2^{-\rho(t)} \cdot d_{z-i}(t) \cdot \operatorname{sgn}^+[e_z(t)] + 1 \right) \right\rfloor, \tag{2.8}$$

$$e_z(t) = 2s_z'(t) - \tilde{s}_z(t). \tag{2.9}$$

$$\rho(t) = \operatorname{clip}\left(\nu_{\min} + \left\lfloor \frac{t - N_x}{t_{\text{inc}}} \right\rfloor, \{\nu_{\min}, \nu_{\max}\} \right) + D - \Omega, \tag{2.10}$$

$$\omega_z^{(i)}(t+1) = \operatorname{clip}\left(\omega_z^{(i)}(t) + \omega_{z,\text{offset}}^{(i)}(t), \{\omega_{min}, \omega_{max}\} \right), \tag{2.11}$$

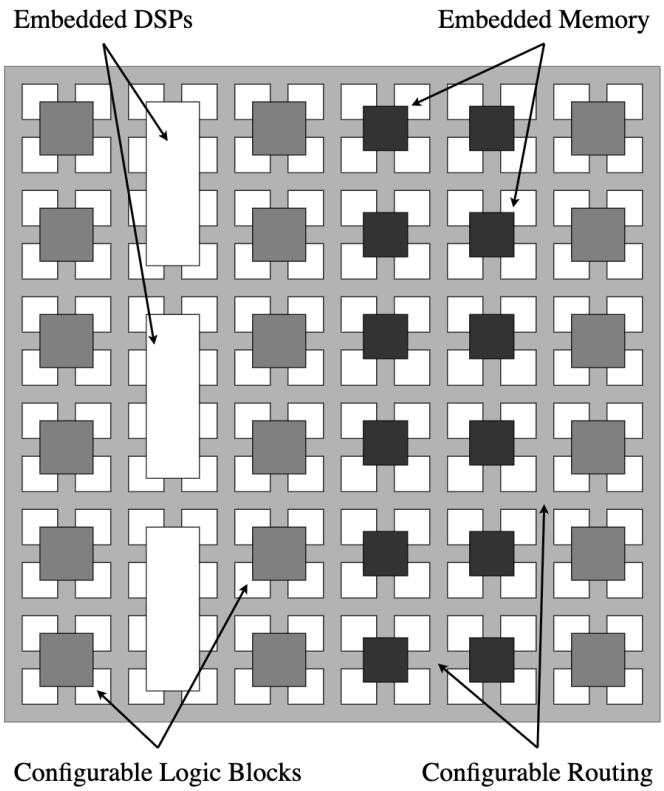
Embedded DSPs
Embedded Memory
Configurable Logic Blocks
Configurable Routing

Figure 2.5

Chapter 3

State-of-the-Art

Table 3.1: The number of processed samples interleaving $z-1$ and $t-1$ dependencies, including the novel FID ordering in addition to the standard orderings.

	BSQ	**BIP**	**BIL**	**FID**
$z-1$	$N_x \cdot N_y$	1	N_x	$N_z - 1$
$t-1$	1	N_z	1	N_z

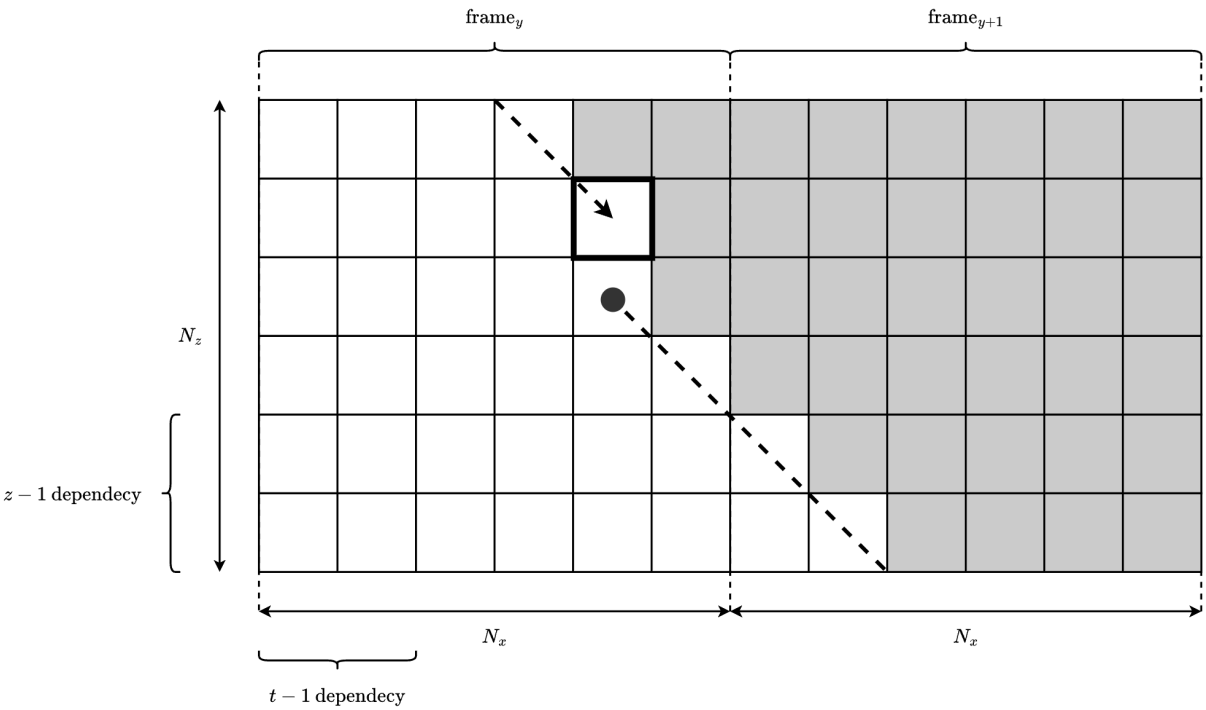

Figure 3.1

$$\operatorname{sgn}^{+}\left[e_z(t)\right] = \begin{cases} -1, & \text{if } |s_z(t) - \hat{s}_z(t)| \leq m_z(t) \text{ and } \tilde{s}_z(t) \text{ is odd,} \\ 1, & \text{if } |s_z(t) - \hat{s}_z(t)| \leq m_z(t) \text{ and } \tilde{s}_z(t) \text{ is even,} \\ \operatorname{sgn}^{+}\left[s_z(t) - \hat{s}_z(t)\right], & \text{otherwise.} \end{cases} \tag{3.1}$$

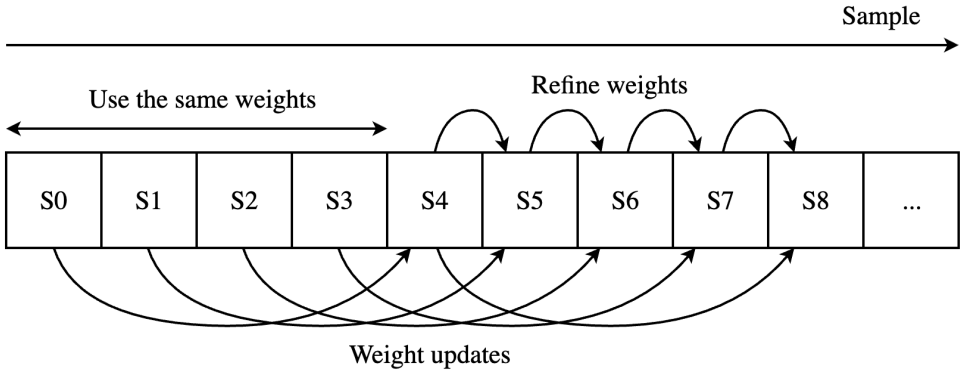

Figure 3.2

$$\omega^{(i)}_{z,\text{unclipped}}(t+1) = \begin{cases} \omega^{(i)}_z(t), & x \leq 2, \\ \omega^{(i)}_z(t-3) + \omega^{(i)}_{z,\text{offset}}(t-3), & x = 3, \\ \left\lfloor \frac{1}{2}\left(\omega^{(i)}_z(t-3) + \omega^{(i)}_{z,\text{offset}}(t-3) + \omega^{(i)}_z(t)\right) \right\rfloor, & \text{otherwise.} \end{cases} \tag{3.2}$$

Chapter 4

Previous Work

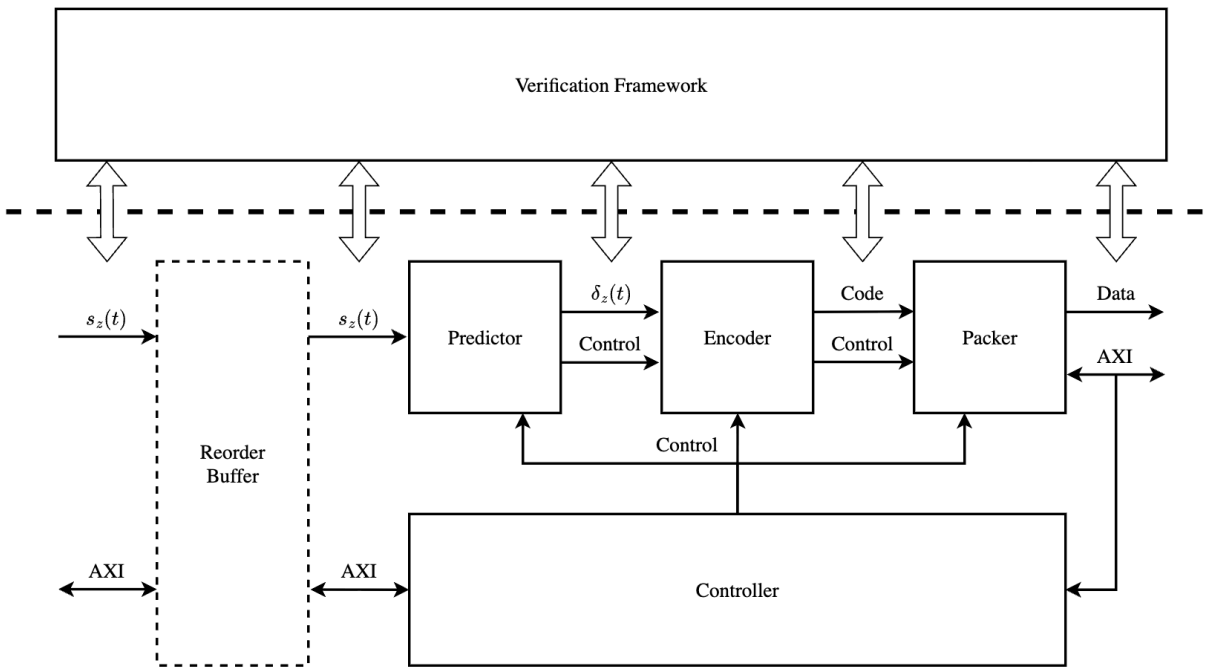

Figure 4.1

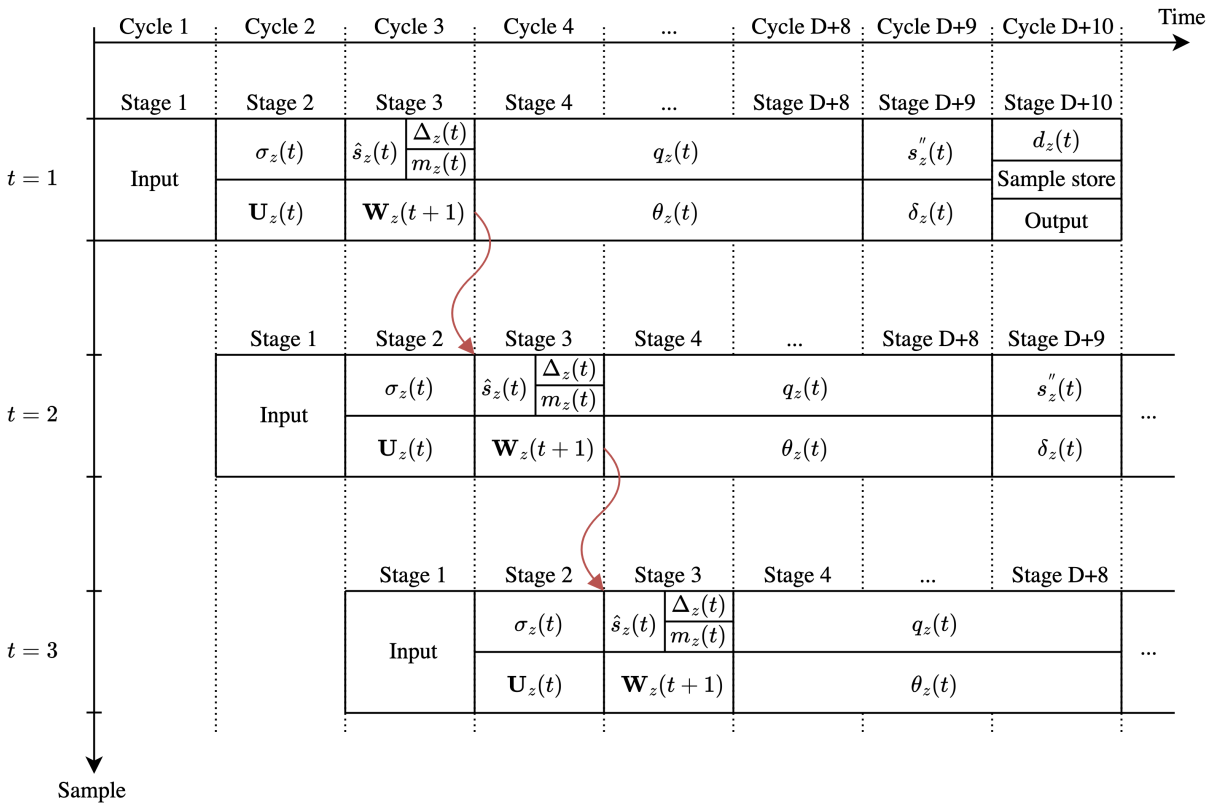
Time
Cycle 1
Cycle 2
Cycle 3
Cycle 4
...
Cycle D+8
Cycle D+9
Cycle D+10
Stage 1
Stage 2
Stage 3
Stage 4
...
Stage D+8
Stage D+9
Stage D+10
$t = 1$
Input
$\sigma_z(t)$
$\hat{s}_z(t)$
$\Delta_z(t)$
$m_z(t)$
$q_z(t)$
$s''_z(t)$
$d_z(t)$
Sample store
$\mathbf{U}_z(t)$
$\mathbf{W}_z(t+1)$
$\theta_z(t)$
$\delta_z(t)$
Output
$t = 2$
$t = 3$
Sample

Figure 4.2

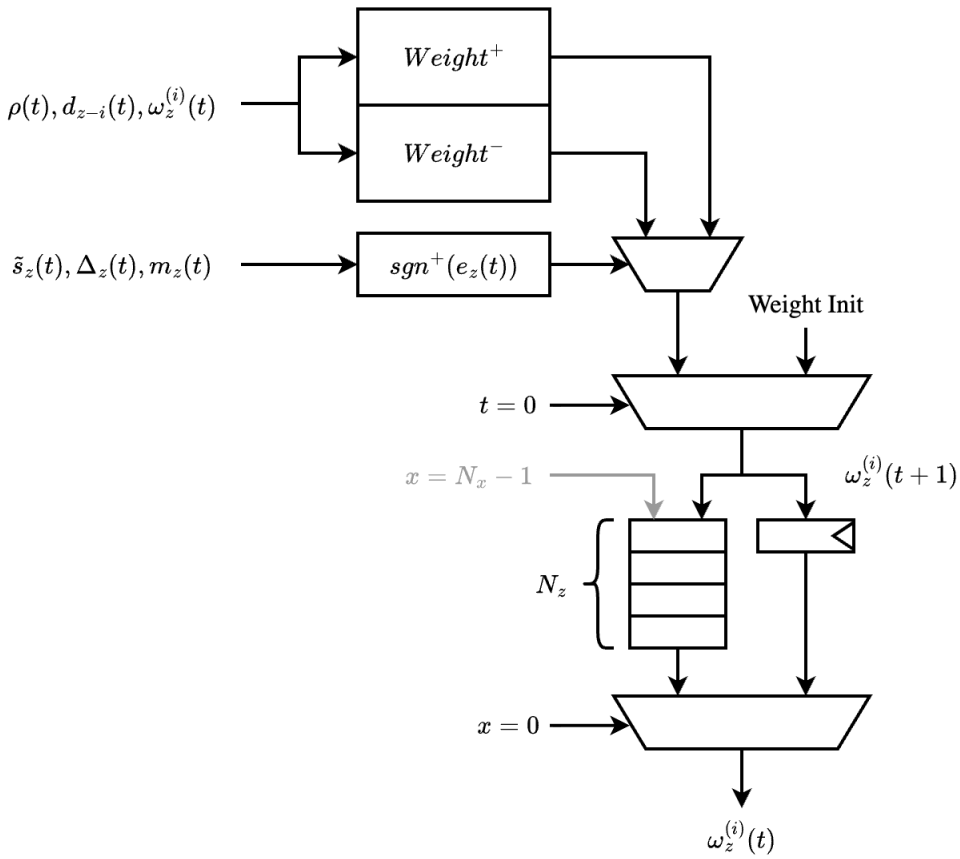
$Weight^+$
$Weight^-$
$\rho(t), d_{z-i}(t), \omega_z^{(i)}(t)$
$\tilde{s}_z(t), \Delta_z(t), m_z(t)$
$sgn^+(e_z(t))$
Weight Init
$t = 0$
$x = N_x - 1$
$\omega_z^{(i)}(t+1)$
N_z
$x = 0$
$\omega_z^{(i)}(t)$

Figure 4.3

Chapter 5

Implementation and Verification

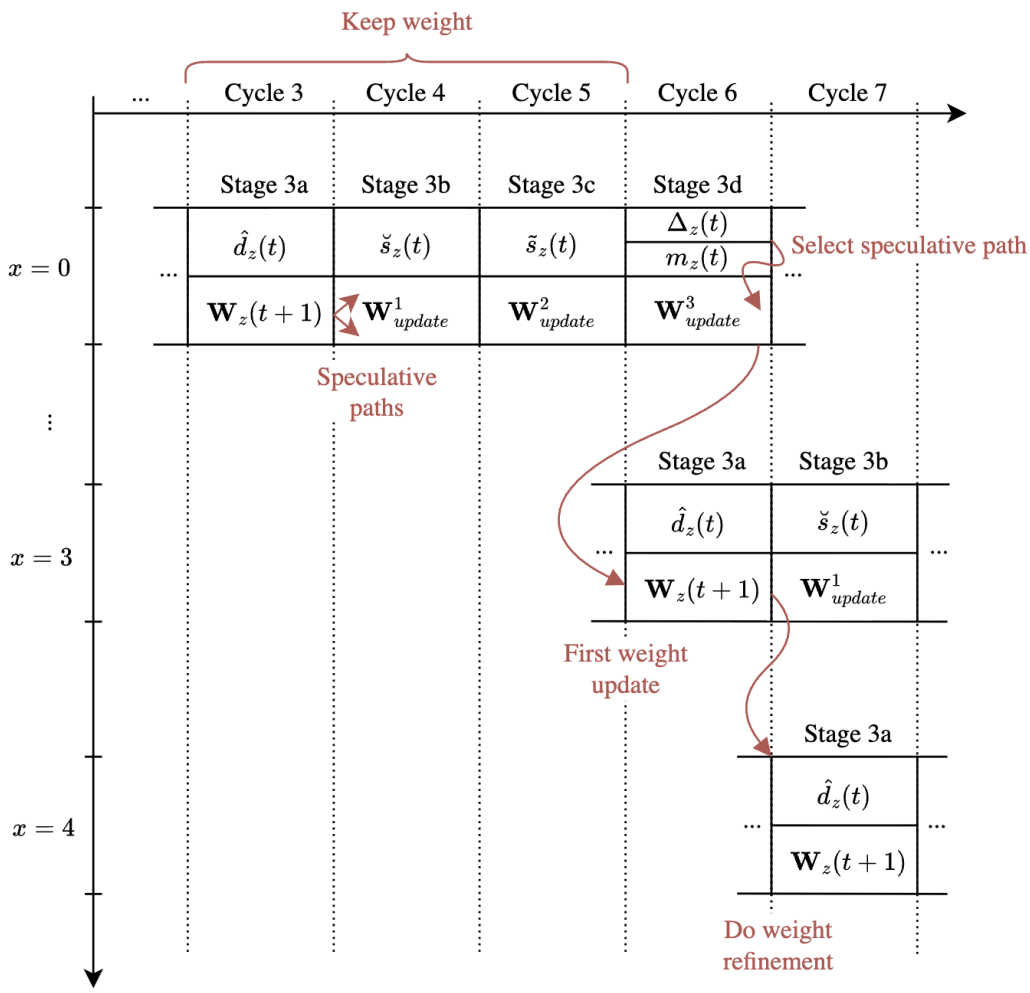

Figure 5.1

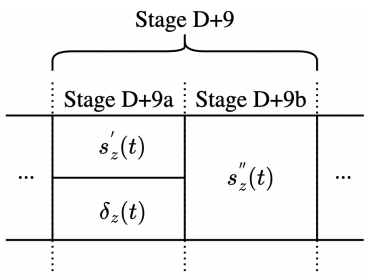

Figure 5.2

Table 5.1: Selected predictor and hybrid encoder parameters. Values are selected based on the work by Blanes et al., and are the same as those used by Vorhaug [1, 2]. Table recreated from [2].

Name	Symbol	Configuration
Sample type		Unsigned
Dynamic range	D	16
Sample encoding order		BIL
Entropy coder type		Hybrid coder
Quantizer method		Absolute
Number of prediction bands	P	3
Register size	R	64
Local sum type		Narrow column-oriented
Prediction mode		Reduced
Weight component resolution	Ω	19
Weight initialization method		Default
Weight update initial parameter	$\nu_{\min}$	-1
Weight update final parameter	$\nu_{\max}$	4
Weight update change interval	t_{inc}	2^6
Weight update exponent offsets	$\zeta_z^{(i)}, \zeta_z^*$	0
Periodic error limit update		Disabled
Band-varying absolute limits		Disabled
Absolute error limit	a_z	*Varied*
Band-varying relative limits		Disabled
Relative error limit	r_z	0
Sample representative resolution	Θ	3
Sample representative offset	ψ_z	7
Band-varying offset		Disabled
Sample representative damping	θ_z	3
Band-varying damping		Disabled
Unary length limit	$U_{\max}$	18
Rescaling counter size	γ^*	6
Initial count exponent	γ_0	1
Accumulator initialization value	$\tilde{\Sigma}_z(0)$	$4 \cdot 2^{\gamma_0}$

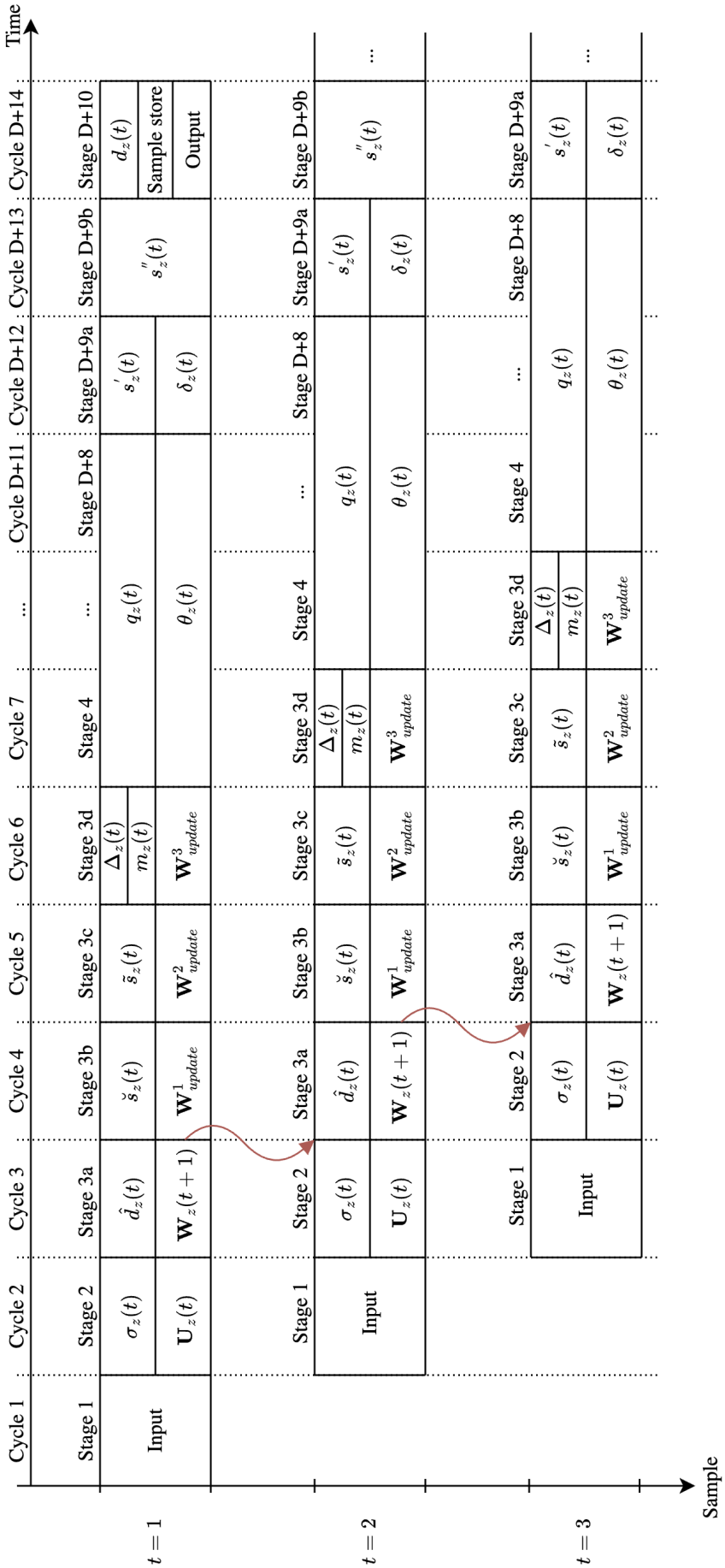
Time
Cycle 1
Cycle 2
Cycle 3
Cycle 4
Cycle 5
Cycle 6
Cycle 7
Cycle D+8
Cycle D+11
Cycle D+12
Cycle D+13
Cycle D+14
Stage 1
Stage 2
Stage 3a
Stage 3b
Stage 3c
Stage 3d
Stage 4
Stage D+8
Stage D+9a
Stage D+9b
Stage D+10
Input
Sample store
Output
Stage 3d
Stage 3c
Stage 3b
Stage 3a
Stage 2
Stage 1
Stage 4
Stage D+9a
Stage D+9b
Stage 3d
Stage 3c
Stage 3b
Stage 3a
Stage 2
Stage 1
Stage 4
Stage D+8
Stage D+9a
Stage D+9b
Sample
$t = 1$
$t = 2$
$t = 3$

Figure 5.3

Chapter 6

Results

Table 6.1: Compression performance, execution time of predictor in parenthesis (TODO: find better way of displaying than in parenthesis)

Image	Predictor	Samples	Time (s)		Throughput (MS/s)	Memory (MB)
Hypso 2	C++	78361920	821	(293)	0.095	32415
Aviris	C++	77643776	1166	(290)	0.066	32091
Landsat	C++	6291456	103	(22)	0.061	2669
Hypso 2	Python	78361920	2818	(2169)	0.027	38996
Landsat	Python	6291456	263	(161)	0.024	3204

Table 6.2: Decompression

Image	Samples	Time (s)	Throughput (MS/s)	Memory (MB)
Hypso 2	78361920	1093	0.072	32414
Aviris	77643776	1834	0.042	32091
Landsat	6291456	149	0.042	2667

Table 6.3: Delayed weight updates, relative 16 and absolute 4

Image	Delayed	Ordinary	Decrease
Hypso 2	2.47	2.49	−0.80%
Aviris	4.31	4.41	−2.23%
Landsat	4.82	5.21	−7.49%

Table 6.4: Reductions in compression performance for coastline image for this work and the work by Jia et al.

PAE	Coastline, Jia et al.	Coastline, this work
0	−1.53%	−0.50%
1	−2.88%	−0.78%
3	−3.86%	−0.63%
7	−4.30%	−0.98%
15	−1.23%	−1.10%

Table 6.5: PSNR of decompressed images with delayed weight updates. Difference compared to without delayed updates are degligable, exact same when rounded to two decimals.

PAE	CR	PSNR (dB)
0	2.02	N/A
1	2.53	225.86
3	3.15	207.94
7	4.03	192.54
15	5.39	178.10

Figure 6.1: Decoded Hypso 2 image

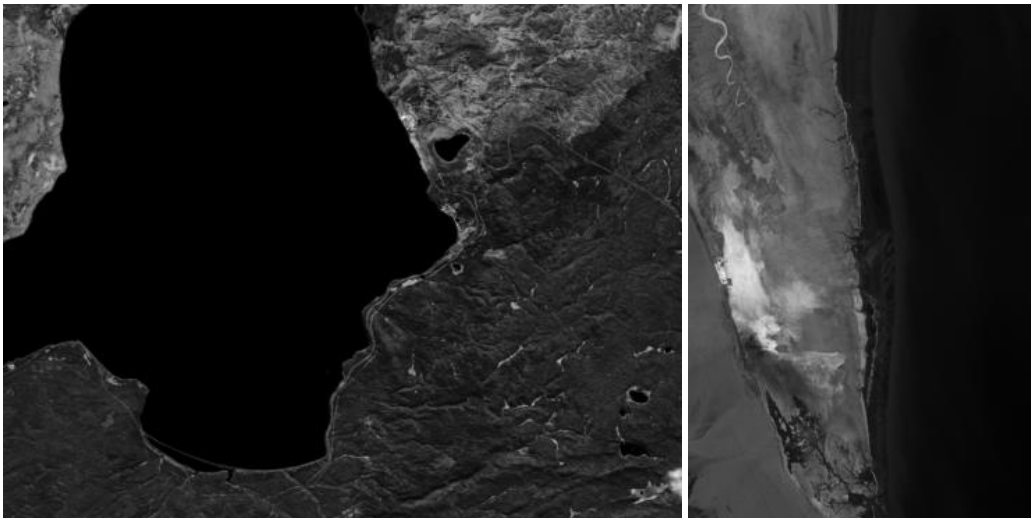

Figure 6.2: Decoded Aviris and Landsat images.

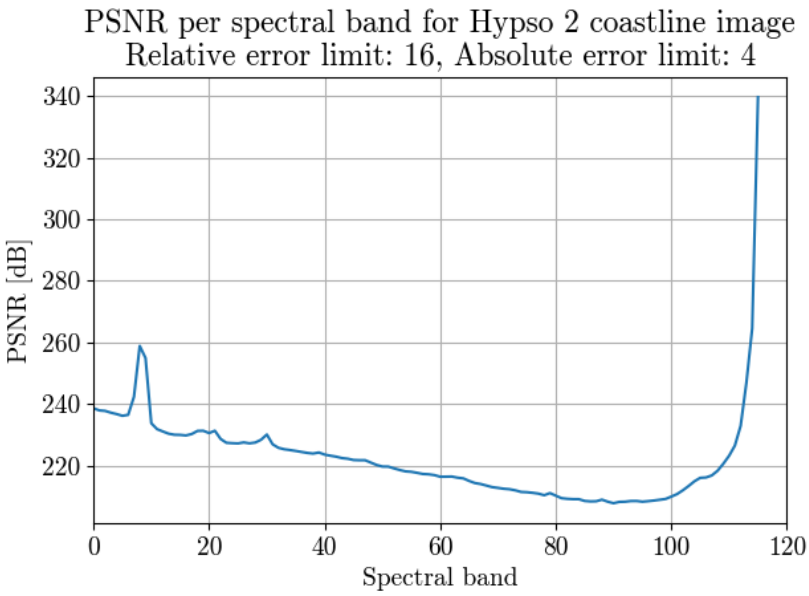

Figure 6.3

Table 6.6

FPGA	LUT	FF	DSP	BRAM	Power	Frequency
Zynq-7030	5958	3425	4	83.5	152 mW	90.9 MHz
Zynq ZU7EV	6331	3442	4	79.5	273 mW	200.0 MHz
Zynq-7030 (predictor)	2841	2212	4	69.5	263 mW	125.0 MHz
Zynq ZU7EV (predictor)	2969	2185	4	65.5	467 mW	312.5 MHz

Chapter 7

Discussion and Further Work

Chapter 8

Conclusions

Bibliography

[1] I. Blanes, A. Kiely, M. Hernández-Cabronero and J. Serra-Sagristà, 'Performance Impact of Parameter Tuning on the CCSDS-123.0-B-2 Low-Complexity Lossless and Near-Lossless Multispectral and Hyperspectral Image Compression Standard,' *Remote Sensing*, vol. 11, no. 11, Jan. 2019, ISSN: 2072-4292. DOI: 10.3390/rs11111390.

[2] D. Vorhaug, 'Hyperspectral Image Compression Accelerator On FPGA Using CCSDS 123.0-B-2,' M.S. thesis, NTNU, 2024.

[3] L. Jia, Q. Wang, L. Zhang, C. Song and P. Zhang, 'Removal of the Feedback Loop in CCSDS 123.0-B-2 during Hardware Implementation,' *IEEE Geoscience and Remote Sensing Letters*, 2025, ISSN: 1545-598X, 1558-0571. DOI: 10.1109/LGRS.2025.3601194.

Appendix A

Block Design

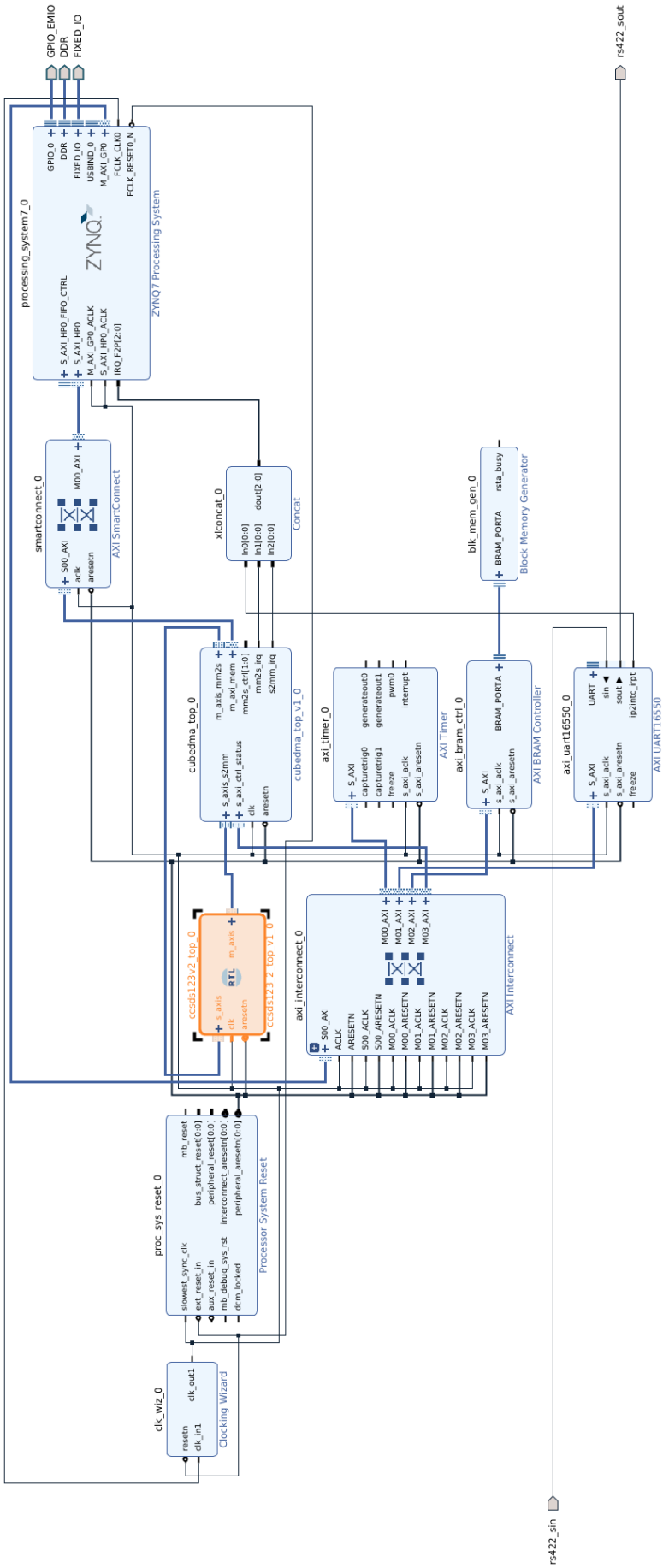
GPIO_EMIO
DDR
FIXED_IO
processing_system7_0
GPIO_0
DDR
FIXED_IO
USBIND_0
M_AXI_GP0
FCLK_CLK0
FCLK_RESET0_N
S_AXI_HP0_FIFO_CTRL
S_AXI_HP0
M_AXI_GP0_ACLK
S_AXI_HP0_ACLK
IRQ_F2P[2:0]
ZYNQ7 Processing System
smartconnect_0
S00_AXI
aclk
aresetn
M00_AXI
AXI SmartConnect
xlconcat_0
In0[0:0]
In1[0:0]
In2[0:0]
dout[2:0]
Concat
blk_mem_gen_0
BRAM_PORTA
rsta_busy
Block Memory Generator
cubedma_top_0
m_axis_mm2s
m_axi_mem
mm2s_ctrl[1:0]
mm2s_irq
s2mm_irq
s_axis_s2mm
s_axi_ctrl_status
clk
aresetn
cubedma_top_v1_0
axi_timer_0
S_AXI
capturetrig0
capturetrig1
freeze
generateout0
generateout1
pwm0
interrupt
s_axi_aclk
s_axi_aresetn
AXI Timer
axi_bram_ctrl_0
S_AXI
s_axi_aclk
s_axi_aresetn
BRAM_PORTA
AXI BRAM Controller
axi_uart16550_0
S_AXI
s_axi_aclk
s_axi_aresetn
freeze
UART
sin
sout
ip2intc_irpt
AXI UART16550
rs422_sout
ccsds123v2_top_0
s_axis
clk
aresetn
m_axis
RTL
ccsds123_2_top_v1_0
axi_interconnect_0
S00_AXI
ACLK
ARESETN
S00_ACLK
S00_ARESETN
M00_ACLK
M00_ARESETN
M01_ACLK
M01_ARESETN
M02_ACLK
M02_ARESETN
M03_ACLK
M03_ARESETN
M00_AXI
M01_AXI
M02_AXI
M03_AXI
AXI Interconnect
proc_sys_reset_0
slowest_sync_clk
ext_reset_in
aux_reset_in
mb_debug_sys_rst
dcm_locked
mb_reset
bus_struct_reset[0:0]
peripheral_reset[0:0]
interconnect_aresetn[0:0]
peripheral_aresetn[0:0]
Processor System Reset
clk_wiz_0
resetn
clk_in1
clk_out1
Clocking Wizard
rs422_sin

Figure A.1